FIITJEE **JEE (Main)-2018**

ALL INDIA TEST SERIES

FULL TEST – IV

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

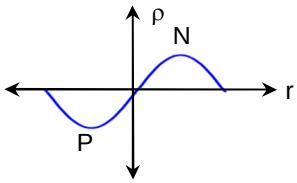
Physics

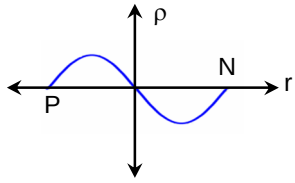
PART – I

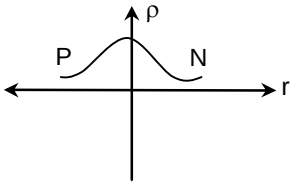
SECTION – A
Straight Objective Type

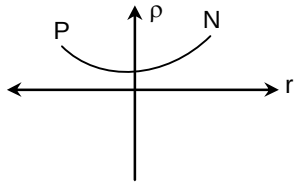
This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A T.V. tower has a height 150 m. The population density around the T.V. tower will be, if the population covered is 50 lacs? (Radius of earth = 6.4×10^6m)
   (A) 708.5 km^{-2} (B) 828.6 km^{-2}
   (C) 856.8 km^{-2} (D) 454 km^{-2}

2. The curve between charge density ρ and distance r near p – N junction will be
   (A)
   

   (B)
   

   (C)
   

   (D)
   

3. The potential of an atom is given by $V = V_o \log\rho(r/r_o)$ where r_o is the constant and r is the radius of the orbits. Assuming Bohr's model to be applicable, which variation of r_n with n is possible (n being principal quantum number).
   (A) $r_n \propto n$ (B) $r_n \propto= \frac{1}{n}$
   (C) $r_n \propto n^2$ (D) $r_n \propto \frac{1}{n^2}$

Space for Rough work

4. Magnets A and B are geometrically similar but the magnetic moment of A is twice that of B. If T_1 and T_2 be the time periods of the oscillation when their poles and unlike poles are kept together respectively, then T_1/T_2 will be

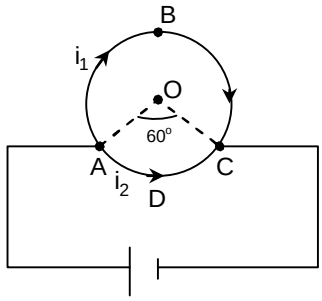
(A) $\frac{1}{3}$ (B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

5. A cell is connected between the points A and C of a circular conductor ABCD of centre O, $\angle AOC = 60^o$. If B_1 and B_2 are the magnitude of magnetic fields at O due to the currents in ABC and ADC respectively, the ratio of B_1/B_2 is.

(A) 5 (B) 1/5

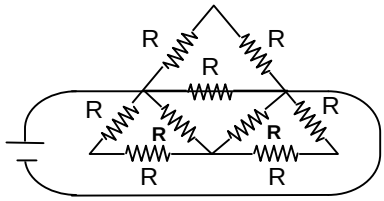
(C) 6 (D) 1



6. For maximum power from battery the internal resistance of battery r is

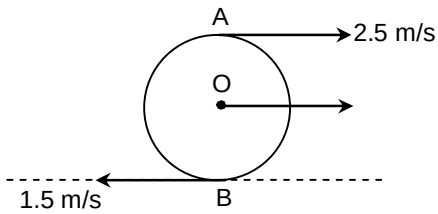
(A) 10 R (B) $\frac{4R}{9}$

(C) $\frac{R}{8}$ (D) $\frac{10R}{9}$



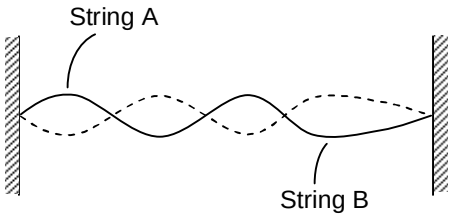
7. A disc having radius r = 2 m is placed on smooth horizontal surface. Velocity of point A and B is given as shown in fig. Distance between point O and axis of rotation at that particular instant is

(A) 1 m
(B) 0.5 m
(C) 0.25 m
(D) 2 m



Space for Rough work

8. Two strings A and B of equal length but un equal linear mass density are tied together with a knot and stretched between two supports. A particular frequency happens to produce a standing wave on each part, with node at the knot as shown. If linear mass density of the two strings A and B are μ_1 and μ_2 respectively, then



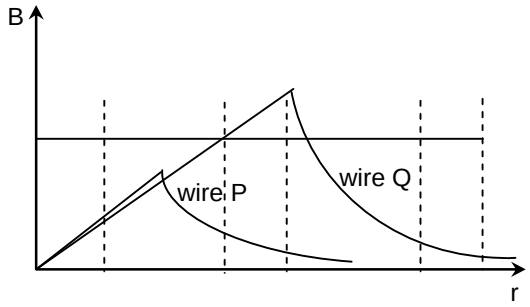
(A) $\mu_1 = \mu_2$

(B) $\mu_1 < \mu_2$

(C) $\mu_1 > \mu_2$

(D) can't be decided

9. Figure shown variation of magnetic field inside and outside two cylindrical wires P and Q carrying currents that are uniformly distributed across the cross section of wires. If current density for two wires are J_P and J_Q then



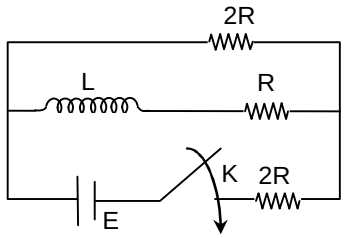
(A) $J_P = J_Q$

(B) $J_P < J_Q$

(C) $J_P > J_Q$

(D) can't be decided

10. Time constant of LR circuit will be



(A) $\frac{L}{R}$

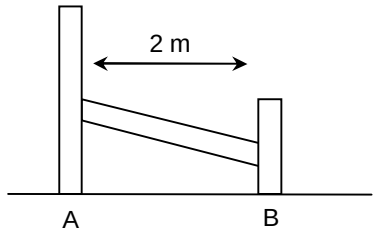
(B) $\frac{2L}{R}$

(C) $\frac{L}{2R}$

(D) None of these

Space for Rough work

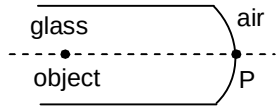
11. Two vertical walls separated by a distance of 2 metres wall A is smooth while wall B is rough with a coefficient of friction $\mu = 0.5$. A uniform rod is propped between them the length of the longest rod that can be probed between the walls is equal to



(A) 2 m (B) $2\sqrt{2}$m

(C) $\sqrt{5}$ m (D) $\frac{\sqrt{17}}{2}$m

12. Radius of curvature of a concave spherical surface separating air-glass medium is R. A point object is placed on the principal axis in the glass $\left(\mu_g = 1.5\right)$. For the image to be real seen by observer in air, distance of the object from P should be



(A) 2 R < u < 3 R (B) u < 2R

(C) u = R (D) u > 3R

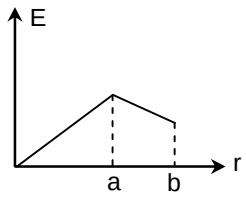
13. A parent radioactive a nucleus A (decay constant D_A) converts into a radioactive nucleus B of decay constant D_B, initially number of atom of B is zero. At any time N_A, N_B are number of atoms of nuclei A and B respectively then maximum value of N_B is

(A) $\frac{D_A N_A}{D_B}$ (B) $\frac{D_B N_A}{D_A}$

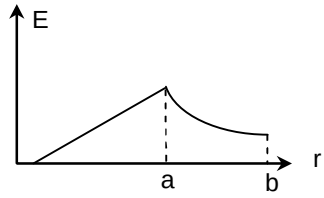
(C) $\frac{(D_A + D_B)N_A}{D_B}$ (D) $\frac{D_B}{(D_A + D_B)}N_A$

14. A capacitor is made of two concentric spherical shells A and B of radii a and b respectively. Where a < b. The external shell B is grounded $(V_B = 0)$ Inner shell i.e$(r \le a)$ is filled with a uniform positive charge of density ρ. Variation of electric field versus distance from the centre.

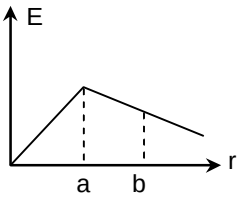
(A)



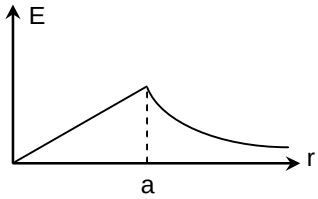
(B)



(C)

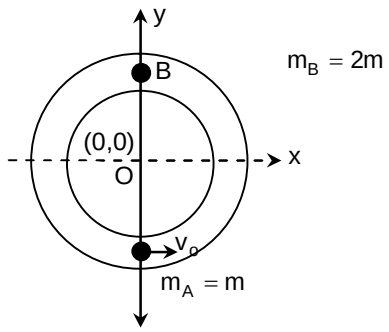


(D)



Space for Rough work

15. A hollow circular tube of radius R and negligible internal diameter is fixed on horizontal surface ball A of mass m is given velocity v_o in the shown direction. It collides with ball B of mass 2 m. collision is perfectly elastic. If centre of loop is origin of co-ordinates system, then co-ordinate of next collision is



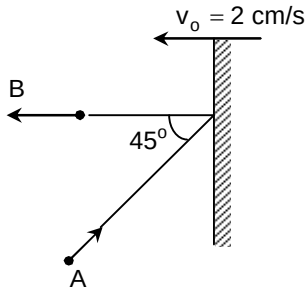
(A) $\left(\frac{\sqrt{3}}{2}R, \frac{-R}{2}\right)$

(B) $\left(\frac{-\sqrt{3}}{2}R, \frac{R}{2}\right)$

(C) $\left(\frac{R}{2}, \frac{-\sqrt{3}}{2}R\right)$

(D) $\left(\frac{R}{2}, \frac{R}{2}\right)$

16. Mirror is moving towards the particle with Speed 2 cm/sec speed of A and B are $10\sqrt{2}$ and 5 cm/sec respectively in the direction shown in figure. Magnitude of velocity of image of the particle B with respect to image of A.



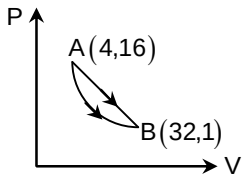
(A) $\sqrt{325}$ cm/sec

(B) 15 cm/sec

(C) 13 cm/sec

(D) $\sqrt{269}$ cm/sec

17. An ideal mono atomic gas is taken through the cyclic process shown in fig. Linear expansion from A to B followed by adiabatic compression back to original state



(A) Efficiency of cyclic process is zero.

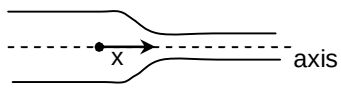
(B) During process A to B heat is released only

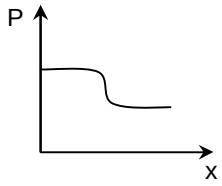
(C) During process A to B heat is absorbed only

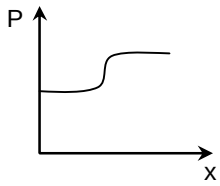
(D) During linear process heat enters and leaves the process

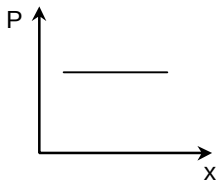
Space for Rough work

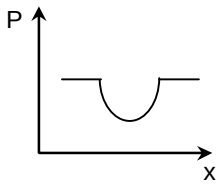
18. An ideal liquid is streamline motion flows through a friction less duct with varying cross section as shown in figure. Pressure P at points along x axis of duct is most likely to be represented by.



(A) 

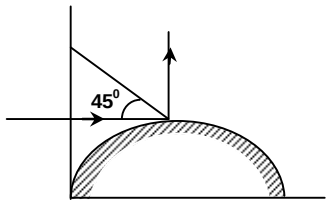
(B) 

(C) 

(D) 

19. If error in the measurement of mass is 0.8 of and in volume it is 0.4% then error in the measurement of density is
(A) 1.2 %
(B) 0.4 %
(C) 0.8 %
(D) 1 %

20. A reflecting surface is represented by the equation $y = \frac{2L}{\pi}\sin\left(\frac{\pi x}{L}\right)$, where $0 \le x \le L$. A ray of light travelling horizontally becomes vertical after reflection with the surface. The co-ordinates of the point where this ray is incident is.



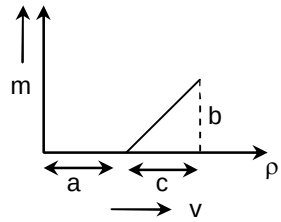
(A) $\left(\frac{L}{4},\frac{\sqrt{2}L}{\pi}\right)$
(B) $\left(\frac{L}{3},\frac{\sqrt{3}L}{\pi}\right)$
(C) $\left(\frac{3L}{4},\frac{\sqrt{2}L}{\pi}\right)$
(D) $\left(\frac{2L}{3},\frac{\sqrt{3}L}{\pi}\right)$

Space for Rough work

21. Kundt's tube can be used to
(A) Produce standing wave
(B) Determine viscosity of water
(C) Produce Doppler's effect in sound
(D) Determine velocity of the source of the disturbance.

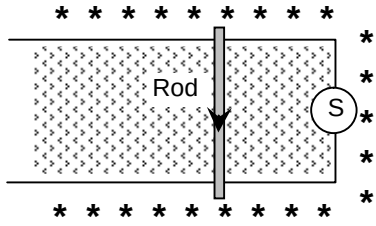
22. The acceleration of electron in the first orbit of hydrogen atom is
(A) $\frac{4\pi^2 m}{h^3}$ (B) $\frac{h^2}{4\pi^2 mr}$
(C) $\frac{h}{4\pi^2 m^2 r^3}$ (D) $\frac{m^2 h^2}{4\pi^2 r^3}$

23. The graph shows how the magnification m produced by a convex thin lens varies with image distance v. What was the focal length of the lens used?
(A) b/c (B) b/ca
(C) bc/a (D) c/b



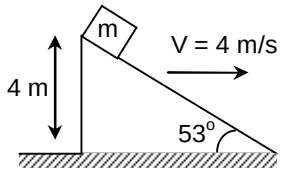
24. Two sound waves of slightly different frequently have amplitude ratio 11/9. What is the different of sound levels in decibels of maximum and minimum intensities head at a point?
(A) 100 (B) 10
(C) 16 (D) 20

25. A conducting rod of mass 50 gm and length 10 cm can slide without friction on two long, horizontal rails. A uniform magnetic field of magnitude 5 MT exists in the region, as shown. A source S is used to maintain a constant current 2A through the rod. If motion of the rod starts from the rest, its speed after 10 sec from the start of the motion will be
(A) 2 cm/s
(B) 8 cm/s
(C) 12 cm/s
(D) 20 cm/s



Space for Rough work

26. An inclined plane is moving with Constant velocity v = 4 m/s on a horizontal surface as shown in figure. If a block of mass 2 kg is kept at top of the incline and their is no friction between the block and the incline, then the distance travelled by the incline till the block reaches bottom of the inclined is $\left(g = 10\ m/s^2\right)$

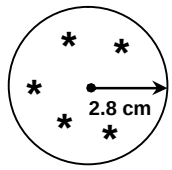


(A) 4 m
(B) 5 m
(C) $1.6\sqrt{5}$ m
(D) $2\sqrt{5}$ m

27. A particle is moving in x-y plane. At certain instant the components of its velocity and acceleration are as follows $V_x = 3m/s$, $V_y = 4m/s$, $a_x = 2\ m/s^2$ and $a_y = 1\ m/s^2$. The rate of change of speed at this moment is.
(A) 4 m/s^2
(B) 2 m/s^2
(C) $\sqrt{3}$ m/s^2
(D) $\sqrt{5}$ m/s^2

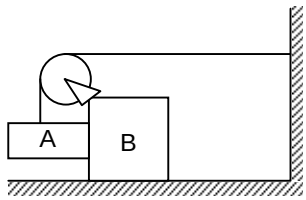
28. In a tug of war, the team that exerts a larger tangential force on the ground wins. Consider the period in which a team is dragging the opposite team by applying a larger tangential force on the ground. Which of the following works are negative?
(A) work by the loosing team on the winning team
(B) work by the ground on the winning team
(C) work by the ground on the loosing team
(D) none of these

29. The magnetic field of a cylindrical magnet that has a pole – face radius 2.8 cm can be varied sinusoidally between minimum value 16.8 T and maximum value 17.2 T at a frequency of $\frac{60}{\pi}$ Hz. Cross section of the magnetic field created by the magnet is shown. At a radial distance of 2 cm from the axis find the amplitude of the electric field (in m N/C) induced by the magnetic field variation.



(A) 140 m N/c
(B) 245 m N/C
(C) 180 m N/C
(D) 240 m N/C

30. The system is released from rest from the position shown in the diagram. If A and B has mass m and M respectively and all surfaces are smooth, then the normal reaction between A and B is



(A) $\frac{2M}{M+m}$
(B) $\frac{Mmg}{2(M+m)}$
(C) $\frac{Mmg}{(M+m)}$
(D) zero

Space for Rough work

Chemistry

PART – II

SECTION – A
Straight Objective Type

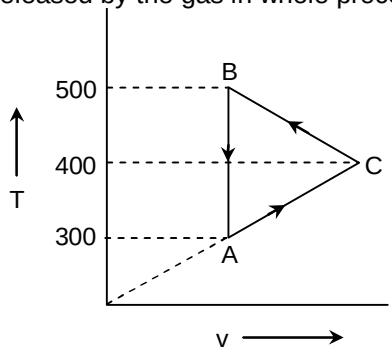
This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Micelles are formed only
(A) Above CMC and above Krafft temperature
(B) Below CMC and below Krafft temperature
(C) Above CMC and below krafft temperature
(D) Below CMC and above krafft temperature

32. On mixing 10 ml of acetone with 40 ml of chloroform the total volume of solution is
(A) < 50 ml (B) > 50 ml
(C) = 50 ml (D) can not predict

33. A Radioactive element A says to B "I am half of what you were when you are one fourth of what I was moreover I was 1.414 time than what you were" If the half life of A is 8 days, what is the half life of 'B' is
(A) 4 days (B) 8 days
(C) 16 days (D) 2 days

34. What will be the work done in process CB for 1 mol of an ideal gas if total 600 cal heat is released by the gas in whole process.



(A) 400 cal (B) 600 cal
(C) 800 cal (D) 1000 cal

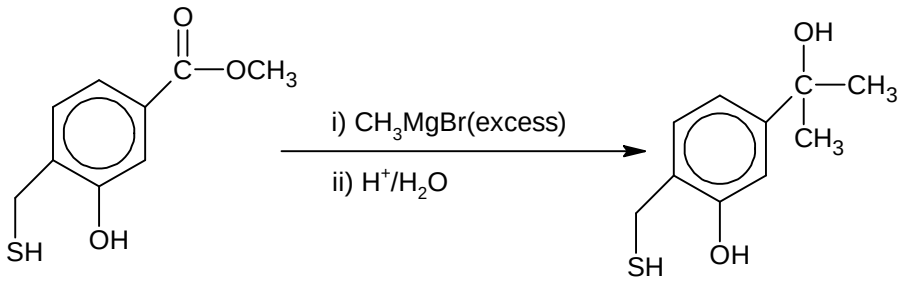
Space for Rough work

35. Calculated E_{cell} for the cell will be:

$$P_t \left| \begin{matrix} H_{2(g)} \\ 1\,atm \end{matrix} \right| \begin{matrix} 1\,N\ KOH_{(aq)} \\ \alpha = 75\% \end{matrix} \Bigg\| \begin{matrix} \frac{N}{10} HCl_{(aq)} \\ \alpha = 90\% \end{matrix} \left| \begin{matrix} H_{2(g)} \\ atm \end{matrix} \right| Pt$$

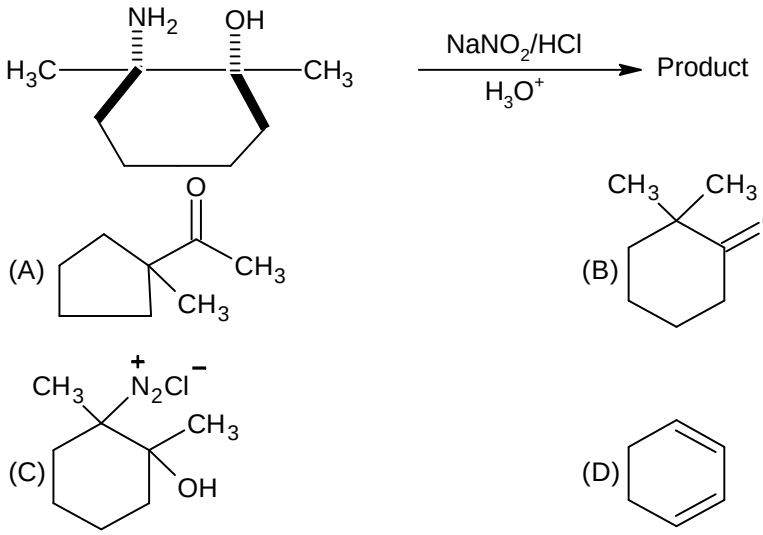
(A) – 0.68 v
(B) + 0.68 v
(C) – 0.378 v
(D) + 0.378 v

36.

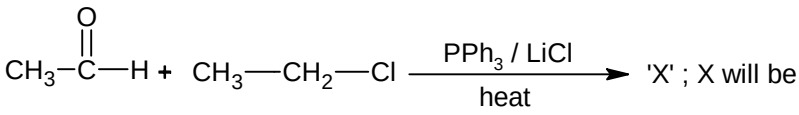


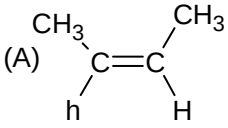
Moles of CH_3MgBr consumed per mol of reactant will be
(A) 4
(B) 2
(C) 6
(D) 3

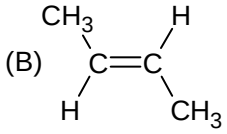
37.



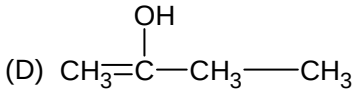
Space for Rough work

38. 

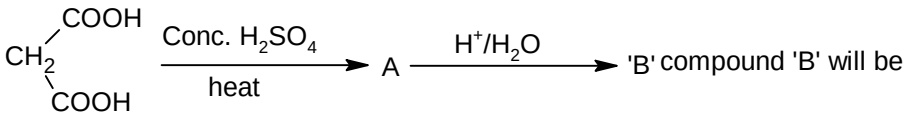
(A) 

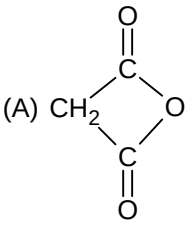
(B) 

(C) $CH_3—CH—O—CH_3$

(D) 

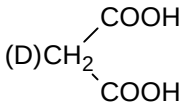
39. Select the incorrect statement for the metallurgy of aluminum:
(A) Bauxite can be enriched by Bayer's process
(B) 'Al' is produced by electro lytic reduction of Al_2O_3
(C) Refining of 'Al' is process by Hoope's electrolysis.
(D) Electrolysis of 'Al' has mixture of $Al_2O_3 + Na_3Alf_6 + CaF_2$ as electrolyte.

40. 

(A) 

(B) $CH_3—COOH$

(C) H_2CO_3

(D) 

Space for Rough work

41. Choose the correct code in which all the molecules have dative π bond.

$I - P_4O_{10}$, $II - (SiH_3)_3N$, $III - P_4O_6$, $IV - BF_3$

(A) I, II, III, IV
(B) II & IV
(C) I, III & IV
(D) I, II & IV

42. When K_2CrO_4 is added to $CuSO_4$ solution, there is formation of $CuCrO_4$ as well as $CuCr_2O_7$. Formation of $CuCr_2O_7$ is due to

(A) Basic nature of $CuSO_4$ solution
(B) Acidic nature of $CuSO_4$ solution
(C) $CuSO_4$ oxidizes CrO_4^{2-} or $Cr_2O_7^{2-}$
(D) It is the typical property of $CuSO_{4s}$

43. Which of the following is low spin to strong field ligands.

(A) $d_{xy}^2 d_{zx}^2 d_{yz}^1$
(B) $d_{xy}^2, d_{yz}^1, d_{zx}^1, d_{x^2-y^2}^1, d_{z^2}^1$
(C) $d_{xy}^2, d_{yz}^1, d_{zx}^1, d_{z^2}^1$
(D) $d_{x^2-y^2}^1, d_{z^2}^1$

44. The configuration of the C – 2 epimer of D – glucose is

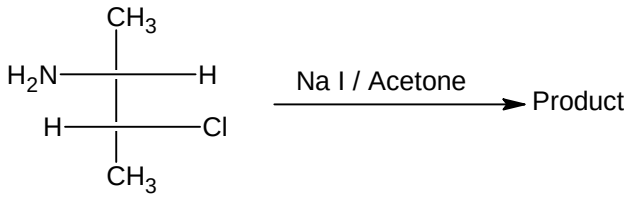
(A) 2 R, 3 S, 4 R, 5 S
(B) 2 S, 3 S, 4 R, 5 R
(C) 3 S, 4 R, 4 S, 5 R
(D) 2 R, 3 S, 4 R, 5 R

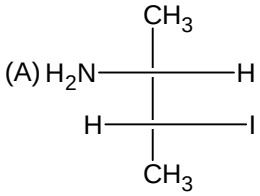
45. $XeF_2 + PF_5 \longrightarrow (A^+B^-)$

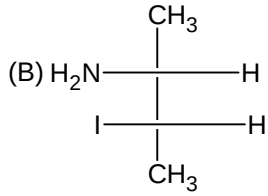
$XeF_4 + 5bF_5 \longrightarrow (C^+D^-)$; choose the correct statement about the equations -

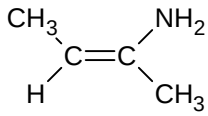
(A) The geometry of A^+ and C^+ is same
(B) The geometry of A^+ and D^- is same
(C) The geometry of B^- and D^- is same
(D) The geometry of B^- and C^- is same

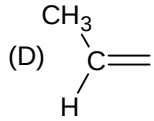
Space for Rough work

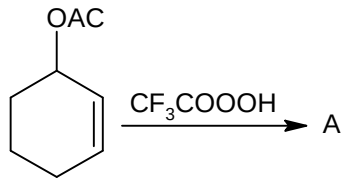
46. 

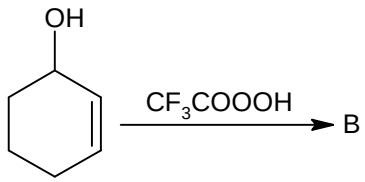
(A) 

(B) 

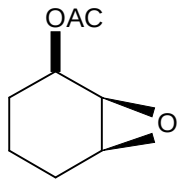
(C) 

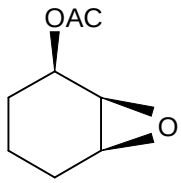
(D) 

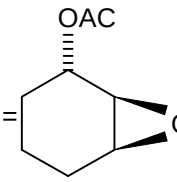
47. 

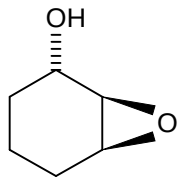


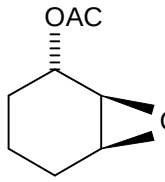
A and B in the above reactions will be:

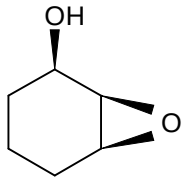
(A) A = 

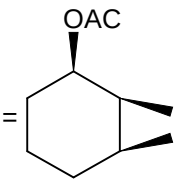
B = 

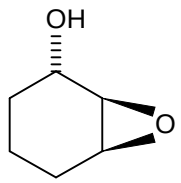
(B) A = 

B = 

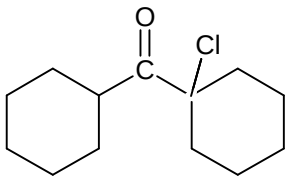
(C) A = 

B = 

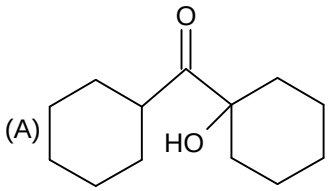
(D) A = 

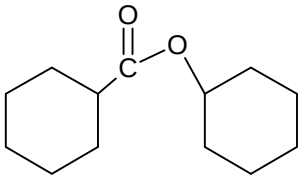
B = 

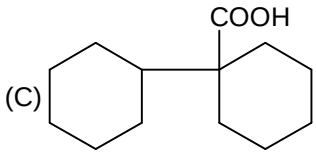
Space for Rough work

48. 

KOH → 'X' ; Product 'X' is -

(A) 

(B) 

(C) 

(D) No reaction

49. Choose the incorrect statement/s among the following:
(1) Then are five heterocyclic bases in DNA & RNA
(2) All the bases except adenine have primary amine grow
(3) Double helix of Nucleic acid is spiraled together by H – bonding
(4) Proteins have no nitrogen content in them.
(A) 1, 2, 4 (B) 2, 3 & 4
(C) 1, 3 (D) 2, 4

50. One gram of charcoal absorbs 100 ml of 0.5 M CH_3COOH to form a mono-layer and thereby the molarity of acetic acid is reduced to 0.49 M. What will be the area of the charcoal absorbed by each molecule of acetic acid?
[given – surface area of charcoal = 3.01×10^2 m^2 / gm]
(A) $5\times10^{-19}m^2$ (B) 5×10^{-20} m^2
(C) 5×10^{-21} m^2 (D) 5×10^{-23} m^2

Space for Rough work

51. $Na_2B_4O_7 + H_2SO_4 \longrightarrow A \xrightarrow{H_2O}$

$\downarrow$

$D \xleftarrow{Na} C$

The incorrect statement regarding substance 'D' is

(A) 'D' will not react with HCl

(B) 'D' will not react with dil H_2SO_4

(C) 'D' will react with conc. H_2SO_4 to form substance 'B'

(D) 'D' will react with conc. H_2SO_4 to form substance 'A'.

52. The energy produced by the de-excitation of $\frac{1}{1000}$ moles of H – atoms from 1st excited to ground state is supplied to 2 moles of an ideal gas. 10 litres of that gas at 27°C are allow to expand reversibly & isothermally. The final volume of the gas will be

(A) 15l (B) -20l

(C) -25l (D) -30l

53. The composition of sample is $F_{0.93}O_{1.00}$. Find the correct code for correct statements.

(i) The %of Fe^{+3} by mass is 11.5%

(ii) The ratio of Fe^{+3};Fe^{+2}ion is 0.17

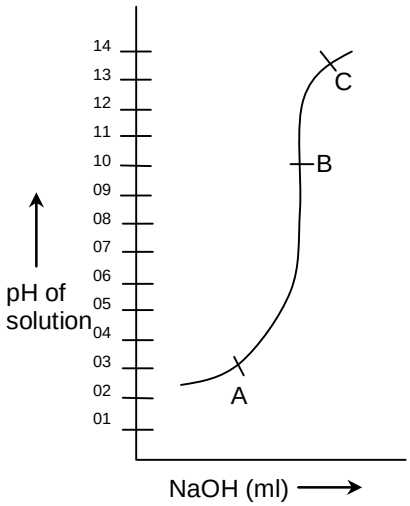
(iii) Fe^{+3} is 7.84 gm.

(A) I, II & III (B) I & II

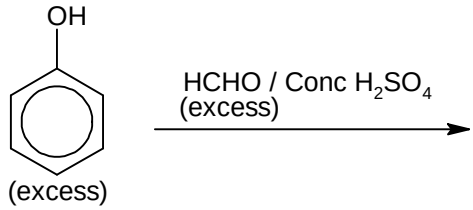
(C) I & III (D) II & III

Space for Rough work

54. A 0.1 M solution of a weak acid HA is titrated by NaOH slowly. The titration curve observed is given. $ka = 4 \times 10^{-5}$ at 25°C. The correct observation for the curve will be:



(A) Solution will make buffer before equivalence point
(B) pH of solution at equivalence point is 5.6
(C) pH of solution at equivalence point is 7
(D) Salt hydrolysis cannot suitable to predict the pH of equivalence point

55. 

'X'. The true statement about product 'X' will be:

(A) Product have alcoholic group
(B) Product is brownish mass with no alcoholic group
(C) The Product has para substitution
(D) The product has ortho substitution

Space for Rough work

56. The position of equilibrium lies to the right in each of these reactions

$N_2H_5^+ + NH_3 \rightleftharpoons NH_4^+ + N_2H_4$

$NH_3 + HBr \rightleftharpoons NH_4^+ + Br^-$

$N_2H_4 + HBr \rightleftharpoons N_2H_5^+ + Br^-$

Based on above information, what is the order of acidic strength?

(A) $HBr > N_2H_5^+ > NH_4^+$ (B) $N_2H_5^+ > N_2H_4 > NH_4^+$

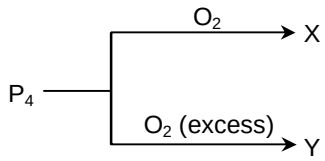
(C) $NH_3 > N_2H_4 > Br^-$ (D) $N_2H_5^+ > HBr > NH_4^+$

57. Considering no electronic repulsion in Helium atom what will be Bohr's energy for both the electrons? Bohr's energy of ground state of H-atom is – 13.6 eV.

(A) – 13.6 eV (B) – 27.2 eV

(C) – 108.8 eV (D) – 54.4 eV

58.



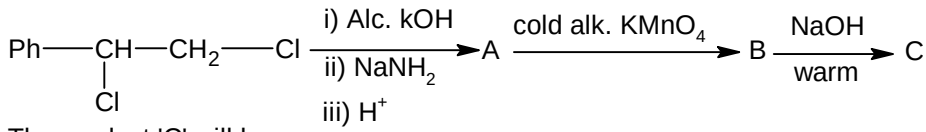
Choose the correct option regarding $d_{p\text{-}o}$ (single bond length)

(A) x < y (B) x > y

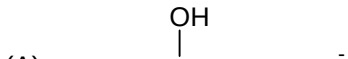
(C) x = y (D) can not compare

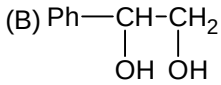
Space for Rough work

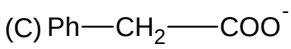
59.

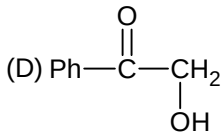


The product 'C' will be

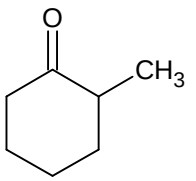
(A) 

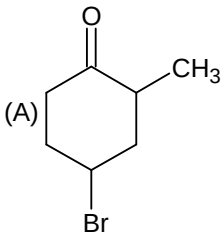
(B) 

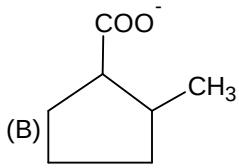
(C) 

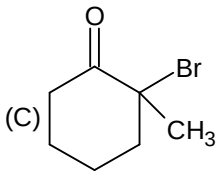
(D) 

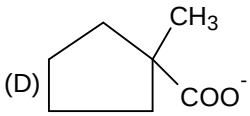
60.



(A) 

(B) 

(C) 

(D) 

Space for Rough work

Mathematics

PART – III

SECTION – A
Straight Objective Type

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If the asymptote of the hyperbola $(x+y+1)^2-(x-y-3)^2=5$ cuts each other at A and the coordinate axis at B and C then radius of circle passing through the points A, B, C is

(A) 3 (B) $\frac{\sqrt{5}}{2}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\sqrt{2}$

62. A variable line $\frac{x}{a}+\frac{y}{b}=1$ moves in such a way that the H.M. of a and b is 8. If c be the least area of triangle made by the line with the coordinate axis then the last digit of c^c is

(A) 2 (B) 4

(C) 1 (D) 6

63. If $\tan^{-1}\alpha=\tan^{-1}(\alpha+k)+\frac{\pi}{4}$ is true for atleast one $\alpha \in [0, 2]$ and $\alpha \in I$, then number of all possible integral values of k is

(A) 0 (B) 1

(C) 2 (D) 3

64. If the angle between the normal chord of parabola which subtends right angle at the vertex is 0 then the maximum value of $f(\alpha)=\sin\theta\cos\alpha+\tan\theta\sin\alpha$ is

(A) $\sqrt{5}$ (B) $\sqrt{3}$

(C) $\frac{4\sqrt{5}}{3}$ (D) 2

Space for Rough work

65. If the straight lines $x + y - 2 = 0, 2x - y + 1 = 0$ and $px + qy - r = 0$ are concurrent then the slope of member of family of lines $2px + 3qy + +4r = 0$ which is farthest from origin is

(A) $-\frac{1}{2}$ (B) -2

(C) $-\frac{2}{3}$ (D) $-\frac{3}{10}$

66. Let $a = \cos^{-1}\cos 20, b = \cos^{-1}\cos 30$ and $c = \sin^{-1}\sin(a + b)$ then maximum value of $\sin(2(a + b + c)x) + \cos^2((a + b + c)x)$ is

(A) $\frac{\sqrt{5}+1}{2}$ (B) $\frac{\sqrt{5}-1}{2}$

(C) $\frac{\sqrt{3}-1}{2}$ (D) none of these

67. A chord AB which bisected at (1, 1) is drawn to the hyperbola $7x^2 + 8xy - y^2 - 4 = 0$ with centre C. which intersects its asymptotes in E and F. If equation of circumcircle of ΔCEF is $x^2 + y^2 - ax - by + c = 0$, then value of $\frac{23(a - b + c)}{12}$ is equal to

(A) 1 (B) 2
(C) 3 (D) 4

68. If normal at (5, 3) of hyperbola xy – y – 2x – 2 = 0 intersects the curve again at $(\alpha, \beta - 29)$, then $\frac{\beta}{\alpha}$ is

(A) 10 (B) 20
(C) 30 (D) 40

69. If the shortest distance between $2y^2 - 2x + 1 = 0$ and $2x^2 - 2y + 1 = 0$ is d then the number of solution of $|\sin\alpha| = 2\sqrt{2}d(\alpha \in [-\pi, 2\pi])$ is

(A) 3 (B) 4
(C) 5 (D) none of these

Space for Rough work

70. From a point perpendicular tangents are drawn to the ellipse $x^2 + 2y^2 = 2$. The chord of contact touches a circle which is concentric with the given ellipse, the ratio of maximum and minimum area of circle is

(A) 3 (B) 4
(C) 5 (D) none of these

71. Vertices of a variable acute angle triangle ABC lies on a fixed circle. If a, b, c are length of sides and angles of triangle ABC and x_1, x_2, x_3 are distances of orthocenter from A, B, C respectively then find the maximum value of $\sqrt{3}\left(\frac{dx_1}{da} + \frac{dx_2}{db} + \frac{dx_3}{dc}\right) + 11$

(A) 1 (B) 2
(C) 3 (D) none of these

72. Let x and y represent the sum and product of two sides of a triangle such that $x^2 = y + z^2$ where z is the third side then Inradius of the triangle is

(A) $\frac{y}{2(z+x)}$ (B) $\frac{z}{2(x+y)}$

(C) $\frac{y\sqrt{3}}{x+y}$ (D) $\frac{z\sqrt{3}}{x+y}$

73. If ellipse E_n is drawn such that it touches ellipse E_{n-1} at the extremities of its major axis, and to have its foci at the extremities of the minor axis of E_{n-1}. If eccentricity e_n of ellipse E_n is independent of n, then the eccentricity of ellipse E_{n-3} is

(A) $\frac{\sqrt{3}-1}{2}$ (B) $\frac{2-\sqrt{3}}{2}$

(C) $\frac{\sqrt{5}-1}{2}$ (D) $\frac{3-\sqrt{5}}{2}$

74. The value of $\left(\sqrt{3}+\tan 1^\circ\right)\left(\sqrt{3}+\tan 2^\circ\right)\left(\sqrt{3}+\tan 3^\circ\right)....\left(\sqrt{3}+\tan 28^\circ\right)\left(\sqrt{3}+\tan 29^\circ\right)$ is

(A) 2^{28} (B) $2^{15}+2-\sqrt{3}$
(C) 2^{29} (D) $2^{14}+2-\sqrt{3}$

Space for Rough work

75. Let $f(x)=1-x+x^2-x^3+....+x^{16}-x^{17}=\alpha_0+\alpha_1(1+x)+\alpha_2(1+x)^2+....+\alpha_{17}(1+x)^{17}$ then α_2 is

(A) 204 (B) 408
(C) 1632 (D) 816

76. Let f(x) be a periodic function such that: $f(x+1)f(x)+f(x+1)+1=f(x),\forall x\in R$ then

(A) $f(x+2)=f(x)$ (B) $f(x+4)=f(x)$
(C) $f(x+6)=f(x)$ (D) $f(x+10)=f(x)$

77. If $xf'(x)-2f(x)=x^4f^2(x),\forall x>0,f(1)=-6,$ then $f\left(\sqrt{2}\right)$ is

(A) $-\frac{3}{2}$ (B) $\frac{1}{2}$
(C) $\frac{3}{2}$ (D) $-\frac{1}{2}$

78. For all real $\alpha>0$, $\int_0^{2\alpha}\frac{\sqrt[4]{\sin(3\alpha-x)}}{\sqrt[4]{\sin(3\alpha-x)}+\sqrt[4]{\sin(\alpha+x)}}dx$ equals

(A) $\frac{\alpha}{2}$ (B) α
(C) $\frac{\alpha}{4}$ (D) 2α

79. In the ΔABC, given $AB=c, AC=b$ and BC=a. Such that: $\frac{2a}{\tan A}=\frac{b}{\tan B}$. The value of $\frac{\sin^2 A-\sin^2 B}{\cos^2 A+\cos^2 B}$ is

(A) 1 (B) $\frac{4}{5}$
(C) $\frac{2}{5}$ (D) $\frac{3}{5}$

Space for Rough work

80. Let z satisfies $|\arg z| < \frac{\pi}{2}$ and $3\bar{z} + 2|z + 2i| + 9i = 0$ then real part of Inz is

(A) $\sqrt{20}$ (B) In $\sqrt{20}$
(C) In $\sqrt{29}$ (D) In $\sqrt{40}$

81. A square matrix P satisfies $P^2 = I - P$ where I is identify matrix. If $P^n = 51 - 8P$ then n is
(A) 4 (B) 5
(C) 6 (D) 7

82. The term independent of x in $\left(1 + x + x^{-2} + x^{-3}\right)^{10}$ is n then last digit of $(n+2)^n$ is
(A) 1 (B) 3
(C) 7 (D) 9

83. Different words are formed using the letters of word FIITJEE, then the probability that the word contains all consonants together is
(A) $\frac{1}{6}$ (B) $\frac{1}{7}$
(C) $\frac{1}{4}$ (D) none of these

84. Let a, b > 0 and $\bar{\alpha} = \frac{\hat{i}}{a} + \frac{4\hat{j}}{b} + b\hat{k}$ and $\bar{\beta} = b\hat{i} + a\hat{j} + \frac{1}{b}\hat{k}$, then the maximum value of $\frac{10}{5 + \bar{\alpha}.\bar{\beta}}$ is
(A) 1 (B) 2
(C) 4 (D) 8

85. If A denotes the area bounded by $f(x) = \left|\frac{\sin x + \cos x}{x}\right|$, x – asis, $x = \pi$ and $x = 3\pi$, then
(A) 1 < A < 2 (B) 0 < A < 2
(C) 2 < A < 3 (D) none of these

86. If $f(x) = (x-1)\left|x^2 - 2007x + 2010\right| + \sin(x-1).\ln x$ has
(A) local min at x = 1 (B) local max at = 1
(C) no local extremum at x = 1 (D) none of these

Space for Rough work

87. If $f(x) \geq 0 \forall\ x \in (0, 2)$ and $y = f(x)$ makes positive intercepts of 2 and 1 unit on x and y-axis respectively and encloses an area of $\frac{3}{4}$ units with axes, then $\int_0^2 xf'(x)dx$ is

(A) $\frac{3}{4}$ (B) 1

(C) $\frac{5}{4}$ (D) $-\frac{3}{4}$

88. Let $x_1 = \tan^{-1}\frac{2}{\sqrt{3}} > x_2 > x_3 \ldots$ are positive real numbers satisfying $\sin(x_{n+1} - x_n) + 3^{\frac{1-2n}{2}} \sin x_n \operatorname{Sin} x_{n+1} = 0$ for $n \geq 1$ then $\lim_{x\to\infty} x_n$ equals

(A) $\frac{-\pi}{8}$ (B) $\frac{\pi}{6}$

(C) $\frac{\pi}{10}$ (D) $\frac{\pi}{12}$

89. The area of the region including between the regions satisfying $\min(|x|, |y|) \geq 1$ and $x^2 + y^2 \leq 5$ is

(A) $\frac{5}{2}\left(\sin^{-1}\frac{2}{\sqrt{5}} - \sin^{-1}\frac{1}{\sqrt{5}}\right) - 4$ (B) $10\left(\sin^{-1}\frac{2}{\sqrt{5}} - \sin^{-1}\frac{1}{\sqrt{5}}\right) - 4$

(C) $\frac{2}{5}\left(\sin^{-1}\frac{2}{\sqrt{5}} - \sin^{-1}\frac{1}{\sqrt{5}}\right) - 4$ (D) $15\left(\sin^{-1}\frac{2}{\sqrt{5}} - \sin^{-1}\frac{1}{\sqrt{5}}\right) - 4$

90. The integer $\int_{\pi/4}^{\pi/2} e^x\ (\ell n(\cos x))(\cos x \sin x - 1)\cos ec^2 x dx$ is

(A) $e^{\pi/4}\left(e^{\frac{\pi}{4}} + \frac{1}{2}\ell n2 - 1\right)$ (B) $e^{\pi/4}\left(e^{\frac{\pi}{2}} + \frac{1}{2}\ell n2 - 2\right)$

(C) $e^{\frac{\pi}{4}}\left(e^{\frac{\pi}{4}} + \frac{1}{2}\ell n2 - 1\right)$ (D) $ee^{\pi/4}\left(e^{\frac{\pi}{4}} + \frac{1}{2}\ell n2 - 1\right)$

Space for Rough work